

swing and the swinging light will cast realistic, moving shadows all over the place. It all comes together to form a more immersive picture...

...Which brings us back to *DOOM III*, where realistic physics is part of the package—you can interact with everything in-game, knock boxes over and break stuff. And the same kind of interactivity extends to the bad guys as well. As John Carmack puts it in an MSNBC.COM interview: "That's one of the things that will give this game fear. If you were backed up to a certain kind of wall in the old games, you knew that nothing was going to happen because the wall was obviously part of the environment. That level of security is not going to be there [with *Doom III*] because the entire world can change and you will not be in command of your environment." So monsters can break through walls, glass windows and even doors to come and get you. Where you gonna hide? Why behind the barrel of my gun, of course! Since a bullet is also simulated by the physics engine, it will have a specific effect based on where you hit an enemy—if you shoot him in the shoulder, he will start spinning or turning backwards or if you decide to aim at his legs, he'll flip over.

But the game that has everyone jumping in joy is the racing phenomenon, *Gran Turismo 3* for Sony's Playstation 2 console. Not only does it feature cars and tracks that look like their real world counterparts, but it handles the physics side so well that race car drivers have been known to use the game to improve their racing skills!

Doing it with intelligence

Physics, pretty graphics and cool lighting



Current technology has allowed developers to offload graphics processing to the video cards, allowing for better simulation of physics in games such as *IL-2: Sturmovik* (right) and *Halo* (above)



Unreal Engine as an Architectural Tool

A major challenge faced by architects is that of presenting their creations to prospective clients. More often than not, 2D sketches are insufficient for communicating the architects' vision, and they have to rely on physical modelling tools or CAD walkthroughs. Both these methods are time-consuming and do not offer ease of modification. Modern 3D game engines have come to the rescue, the fore-runner being Epic Games' Unreal Engine. The engine offers a great tool to developers of gaming and non-gaming products alike in the form of a mouse-driven level editor—UnrealEd. Architects use the tool to create a high polygonal version of their design, apply photo-realistic textures, and use the generated content to present a real-time



3D walkthrough to their clients. Given the networking capabilities of the engine, the client can be on the other side of the country. And modifications can also happen in real-time, since the tool is just a front-end for the game engine, which concurrently updates changes.

effects are all well and good, but when you see an in-game character walk right into a wall or unrealistically overtake you in his inferior car, you are immediately reminded that you are inside the make believe world of games. Well, AI (Artificial Intelligence), as it is called, is perhaps the biggest misnomer in the gaming industry. Things might seem impressive when the soldier that you are shooting at ducks behind a crate and lobs a grenade at you, or as you are crawling through a pipe someone opens the hatch door and drops in an explosive, but when the same sequence is repeated time and again, you know that this isn't some advanced programming magic but some basic (albeit, clever) scripting.

Lionhead Studio's *Black and White* made an honest attempt to rectify the situation. Casting you as a God in an unspoiled Eden, you could control one of several creatures that would observe everything you did, learn, remember and eventually form a personality closely mirroring your actions. Lionhead and its affiliate studios plan to take the technology further in games such as *Project Ego* and *Dimitri*.

Another title to watch is Elixir Studio's *Republic: The Revolution*. Revolutionary AI techniques will allow 'live' citizens to inhabit the dozens of towns and cities in the in-game country of Novistrana. Each person will have detailed daily routines

that range from going to work, to collecting their kids from school, shopping and voting in an election. Each individual will also have their own set of beliefs, emotions, skills and loyalties that govern behaviour, producing consistent and realistic reactions to situations that arise in the game.

Are we there yet?

A while back, nVidia demonstrated the power of its then-latest baby (the GeForce 3 video card) by rendering the computer generated movie *Final Fantasy: The Spirits Within*, in real-time. "Time has come", they said, "the power to deliver a movie-quality game is here." Apparently, developers were listening. At E3, held in Los Angeles in May 2002, a recurring question that was asked at the *DOOM III* booth was: "Why are you rendering this [*DOOM III*]?", only to be answered by, "We're not rendering it... that's the game."

A few years ago, a director used UnrealEd (then shipping with *Unreal Tournament*) to generate 3D versions of the sets for his movie, granting him the ability to preview camera moves by flying through a real-time version of the virtual set. This gave him a pretty good idea of what the set was going to look like before physically building it for the actual shoot. The director was Steven Spielberg, working on the epic *A.I. (Artificial Intelligence)*. That was then. The tools are better now, so is the technology and the hardware driving it. Maybe next time Spielberg will not need the real sets. ■

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